



Original Paper

Climate Change Impacts on Groundwater Resources in the Coastal Karstic Adriatic Area: A Case Study from the Dinaric Karst

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The Bokanjac–Poličnik system, as a complex set of mutually interrelated Dinaric karst catchments and sub-catchments, is a highly vulnerable and limited groundwater source for the wider Zadar area in northern Dalmatia, Croatia. Based on hydrogeological, hydro-chemical, and hydrological research, including the prediction of groundwater discharge by the end of the twenty-first century, a complex study was performed with the following main aims: (1) groundwater protection in the present state, (2) assessment of future groundwater protection, and (3) prediction of drinking water availability and quality under the predicted climate change conditions. Long-term prediction of changes in groundwater quantity, as well as investigations of trends in groundwater quality, will allow us to protect this essential natural resource with respect to possible negative trends. The results showed that a significant decrease in the quantity of available groundwater is possible and that extraction will have to be well planned because any decrease in the groundwater pressure in this area will cause a further decrease in quality, especially regarding the possibility of seawater intrusions into the aquifer. The results of this study were incorporated in sanitary protection zones.

KEY WORDS: Karst groundwater, Coastal aquifer, Climate change, Seawater intrusion, Groundwater quantity and quality.

INTRODUCTION

The coastal areas and their associated water resources in the Mediterranean region are often exposed to the greatest pressures. Almost one-quarter of the world's population lives in regions where groundwater is used up faster than it can be replenished (Gleeson et al., 2012). In the research area, this depletion is partially triggered by anthropogenic impacts (hydrotechnical interventions, in-

creased use of groundwater for the water supply and agriculture, and pressures on water quality) and by natural causes, especially in the context of predicted climate changes. While anthropogenic pressures can be reduced and partly compensated for on a regional scale, global climate changes become an unavoidable threat to the sustainability of coastal water systems, with increasing occurrences of seawater intrusions, especially in the Mediterranean region (including the Adriatic area) where coastal karst aquifers are highly threatened (Fritz 1978; Custodio and Bruggeman 1982; Custodio 2002; Bolle 2003; Terzić 2004; Pavičić et al. 2006; Terzić et al. 2007, 2010; Borović et al. 2019).

Climate changes affect many spheres of life, and one of the most important is the supply of drinking

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water (Green et al. 2011; Simonffy 2012; Taylor et al. 2012; Collet et al. 2015). Groundwater is affected by changes in the major long-term climate variables such as air temperature and precipitation. A greater variability in rainfall could mean more frequent and prolonged periods of high or (mainly) low groundwater levels, while seawater intrusion in coastal aquifers can appear due to sea level rise or decreased groundwater pressure (Stigter et al. 2012; Vera et al. 2012). The fact that karst groundwater is at high risk was confirmed by the results of models which predict the effects of climate change on hydrological cycle factors, particularly influencing available groundwater quantities (Hartmann et al. 2012; Brenner et al. 2018; Chen et al. 2018). The problem is even greater because of the increasing water demand (Loáiciga et al. 2000). Sarrazin et al. (2018) have found that groundwater recharge is not only sensitive to precipitation decrease, but also to land cover properties using the V2Karst large-scale model. High vulnerability of karst aquifers, and high future risk predicted, requires efficient planning and better management of karst groundwater to preserve that vital water resource.

This paper presents a conceptual model for the functioning of the coastal karst aquifer and analyzes several other types of problems, namely (1) relationship among surface water, groundwater, and seawater; (2) groundwater quantities used for water supply; (3) water balances in the present state and according to the predicted climate changes; and (4) assessment of groundwater quality trends. The current approaches to the management of water resources seek to develop different scenarios of possible long-term changes to identify risks in a timely manner and to prepare and optimize protective management measures (Rubinić and Katalinić 2014). Parts of the presented research were published in a conference (Terzić et al. 2014).

STUDY SITE

The research area, i.e., the Bokanjac–Poličnik (BP) system, is situated in coastal karstic terrain in an area called Ravni Kotari, which is part of northern Dalmatia in Croatia (Fig. 1). The central point of the research area is the Bokanjačko Blato karst polje, which represents the largest part of the BP catchment area. The Bokanjačko Blato karst polje was a periodic lake under natural conditions, and since 1901, several public water supply extrac-

tion sites have been situated there (Kaleb et al. 2005). In 1963, this karst polje was drained as part of a broader melioration system design, which included (1) a network of canals within the polje, (2) a tunnel from the polje to the periodic stream Miljašić Jaruga, and (3) a grout curtain on the western side of the polje to prevent further seawater intrusion after the hydraulic head drop. However, after the decrease in the groundwater level in the wider area, during dry hydrological years, seawater intrusions occurred. Additional pressures include intensive agricultural activities in the Bokanjačko Blato and Ravni Kotari karst terrain as well as increased water demand during the summer tourist season, when there is lack of rainfall and, thus, no recharge into the aquifers. Such anthropogenic changes, combined with the more frequent unfavorable climatic conditions, increase the risks of maintaining an adequate water supply from these exploitation sites. To thoroughly understand how karstic systems and aquifers work, how to protect them, and how to predict their behavior with changes in boundary conditions, water resources should be analyzed on the basis of the continuous historical trends of climate and hydrological processes and the consequent changes (Rubinić and Katalinić 2014).

The surface of the BP catchment area is approximately 200 km². The area is situated in the coastal area of northern Dalmatia in Croatia and is a typical karst system with numerous groundwater phenomena. The water supply system extracts spring or groundwater at four extraction sites: the Golubinka coastal karst spring, the Bokanjac pit wells (B-4 and B-5), the estavelle Jezerce, and the Boljkovac karst spring/pit well (Fig. 1). The Golubinka spring is in the northern part of the catchment. Golubinka is a coastal karst spring and a partially submerged cave whose groundwater is periodically slightly brackish due to seawater penetration into the aquifer. The temporary/torrential surface flow in Miljašić Jaruga forms in the eastern part of the catchment, which is composed mostly of low permeable flysch rocks. Morphologically, the catchment area is quite smooth and mild. The relief is “inverted,” i.e., the hills follow synclinal axes, while the valleys follow anticlines. The mean elevation of the terrain is low (up to 100 m a.s.l.). An important fact is that the sea level has changed since the last glaciations (Würm) and is now approximately 100 m higher (Šegota 1982; Masse and Montaggioni 2001; Surić 2009). In the last 6000 years, a stabilization of sea level has taken place (Pirazzoli 2005). Nevertheless, the large

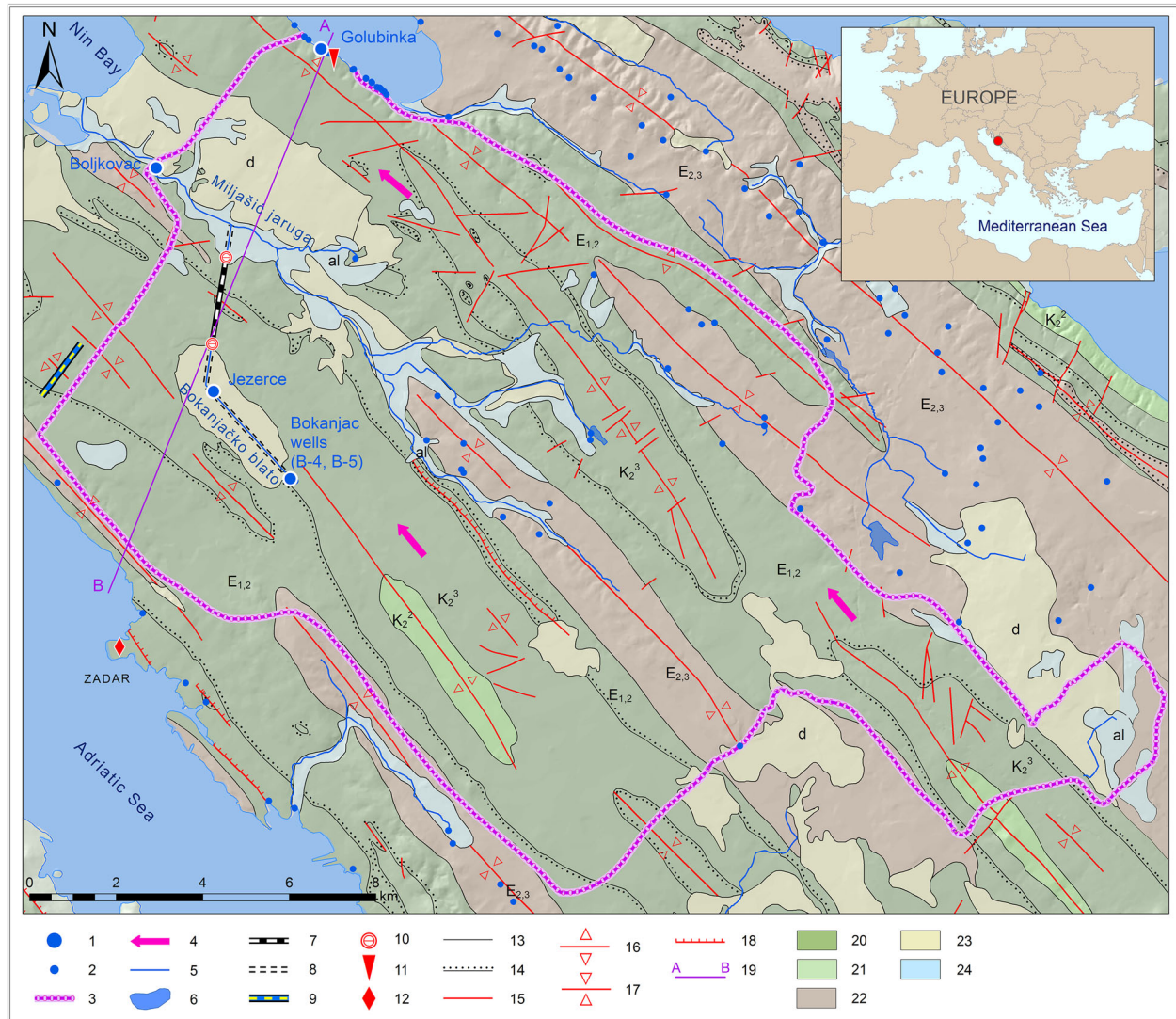


Figure 1. Hydrogeological map of the test area with the situation map. 1, sampling location; 2, springs of different discharges; 3, groundwater divide; 4, groundwater flow direction; 5, stream; 6, pond; 7, tunnel; 8, channel; 9, grout curtain; 10, hydrological station; 11, tide gauge station; 12, climatological station; 13, 14, geological boundaries; 15–18, structural–tectonic features; 19, cross section; 20, carbonate rocks of high permeability; 21, carbonate rocks of medium permeability; 22, clastic rock masses of low permeability; 23, 24, Quaternary deposits. The map shows the position of the cross sections A–B used for the conceptual model presented in Fig. 3.

sea level rise in the Holocene resulted in a much deeper karstification than what would be expected with the recent sea level. Čupić et al. (2011) determined, based on the analysis of the recorded mareographic data for the period of 1993–2009, that the trend of sea level rise in the mid-Adriatic region is $3\text{--}4\text{ mm year}^{-1}$ (in the middle and south Adriatic). These results correspond to the EEA (European Environment Agency) estimations (EEA 2016) based on satellite images from 1992 to 2014. The

estimates are also given in the same document for the period between 2081 and 2100. According to those estimates, an exceptionally large sea level rise of 30–40 cm is expected. The largest part of the BP system area is the Bokanjačko Blato catchment. The water surpluses from Bokanjačko Blato are being drained into Miljašić Jaruga through the drainage tunnel. This drainage has changed not only the water regime of the surface waters but also that of the groundwater, resulting in a decrease in their natural

oscillations. The risk of seawater intrusion into these aquifers has increased, especially under conditions of over-extraction.

Hydrogeological relations are mostly a consequence of the geological circumstances. The whole region is composed of highly permeable karstified rocks (carbonates) and some clastic rocks (flysch) of low permeability (Fig. 1). In recent decades, many manmade interventions in this hydrogeological system were related to extensive agricultural production and the water supply. The BP hydrogeological system, close to the city of Zadar, has an important role in the water supply of Zadar and the whole region. The water demand in this area is high and multiplies during the summer season, when the population dramatically increases due to tourism. The four extraction sites noted above cover more than one-quarter of the total water demand. The total minimum extraction quantity is $0.21 \text{ m}^3 \text{ s}^{-1}$, and the maximum is $1.4 \text{ m}^3 \text{ s}^{-1}$. The natural regime of the karst polje Bokanjačko Blato has changed significantly. The groundwater levels are lower, and over-pumping has caused (slight) seawater intrusion. To prevent this intrusion, a grout curtain in the western part of the catchment was established during the 1970s. This curtain prevented intrusions from that direction, and the groundwater levels in the zone of the extraction sites of Bokanjačko Blato increased 3–4 m, but the other sides of this open karst system remain without such protection. A decrease in recharge, a consequence of climate changes, currently endangers this system. The levels are now lower than ever, and the water quality and quantity of this catchment are seriously at risk.

Land use and management practices have impacts on groundwater quality. A land use map shows the human use of the land, such as settlements, and seminatural habitats, such as arable fields, pastures, and managed woods. Land use is defined as arrangements, activities and inputs that people undertake in a certain land cover type to produce, change, or maintain it (Watson et al. 2011). The actual land use for the BP catchment area was determined based on the Corine Land Cover 2012. The investigated area is mostly covered by forest (50%) and agriculture (41%), which is the most important land use driver. Thus, in addition to the karst nature and seawater intrusion, agricultural activity represents a major impact on water quality.

The Croatian coast is located between the Dinarides and the Adriatic Sea, with the location dictating the climatological characteristics of the

analyzed area, i.e., long, extremely hot, dry summers and significant precipitation in the late autumn, and winter parts of the year. Since the 1960s, the characteristic annual values of climatological and hydrological data present unfavorable trends within the Bokanjačko Blato aquifer (Fig. 2), in accordance with the global climate change predictions. Using the Mann–Kendall significance test (Gilbert 1987) for the analyzed period of 1960–2009, it was calculated that only air temperatures recorded at the Zadar climatological station have significant increasing trend ($p < 0.0001$), with a gradient of $0.28 \text{ mm}/10 \text{ years}$. Nevertheless, trends of the other hydrological indicators were present, even though they were not statistically significant. Annual decrease in precipitation has occurred at the Zadar climatological station ($p = 0.055$), with a gradient of $38 \text{ mm}/10 \text{ years}$. Consequently, the mean annual discharge (MAD) points at decreasing trend ($p = 0.199$) in the main collector of surface water (Miljašić Jaruga stream), measured at the Boljkovac station ($50 \text{ L s}^{-1}/10 \text{ years}$). During the same period,

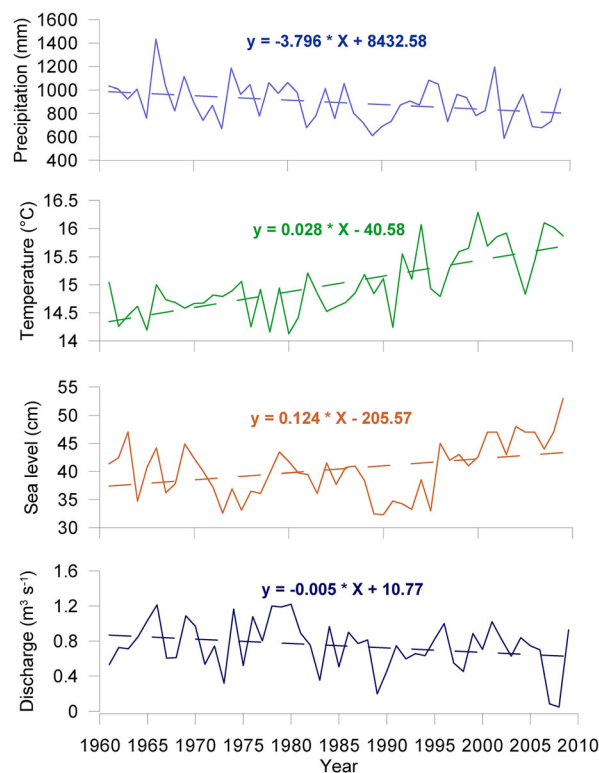


Figure 2. Long-term trends in annual precipitation and mean annual air temperature from the Zadar climatological station and the mean annual discharges from the Boljkovac-Miljašić Jaruga hydrological station (1961–2009).

mareograph Golubinka recorded increasing seawater level trend (0.196 m) with the gradient of 1.2 cm/10 years.

MATERIALS AND METHODS

The hydrological and climatological data were obtained from the hydrological and climatological stations in the catchment area (Fig. 1), and the hydrochemical data were obtained from the springs and wells of the Zadar water supply company, which regularly monitors the quality of water.

The datasets from the Zadar climatological station ($\varphi = 44^{\circ}7'48\text{N}$, $\lambda = 15^{\circ}12'21\text{E}$) cover the period of 1951–2009, while the data from the hydrological station within the catchment (discharges, seawater level) cover the period of 1996–2014. To get insight into longer datasets of the discharge and seawater levels, MAD from the station Boljkovac on Miljašić Jaruga ($\varphi = 44^{\circ}13'53\text{N}$, $\lambda = 15^{\circ}13'0\text{E}$) has been supplemented using regression data from Janković Buk station ($\varphi = 44^{\circ}12'11\text{N}$, $\lambda = 15^{\circ}43'16\text{E}$; 30 km from the Boljkovac station). The main purpose was to improve interpretation of longer series of the discharges and sea level. The data from Golubinka mareograph ($\varphi = 44^{\circ}15'23\text{N}$, $\lambda = 15^{\circ}15'55\text{E}$) have been confirmed using the data from the closest mareograph in Split ($\varphi = 43^{\circ}30'25\text{N}$, $\lambda = 16^{\circ}26'5\text{E}$), 125 km away.

Digital spreadsheets of the average yearly air temperature and precipitation distributions for the relevant 30-year period (1961–1990) were used (Gajić Čapka et al. 2010). Based on the data from the Zadar climatological station, within the CC-WaterS project, the estimation of the mean annual air temperature and the precipitation for the period of 2010–2100 was performed and complemented by the measured data. The influence of the climate changes on the hydrological characteristics was based on two climatological models: RegCM3 (Pal et al. 2007) and Aladin (Bubnová et al. 1995). The Wilcoxon test (Wilcoxon 1945) was performed to assess the homogeneity of the input datasets and results. The nonparametric Mann–Kendall estimate method was used to determine a monotonic increasing or decreasing trend of the variable of interest over time with their statistical significance (Gilbert 1987). The usual significance level of 5% was used (p value).

To evaluate the impacts of seawater intrusions and agricultural activities on water quality, analyses

of nitrate and chloride concentrations trends were performed using single regression analysis. The trends were calculated for the thirteen-year period from 2000 to 2012. The data were not collected at regular intervals throughout the period of observation. To avoid the excessive influence of extreme values and to mitigate the impact of the irregular sampling intervals, the median of measured annual trends was considered as plausible.

Hydrological methodology was based on two critical elements—estimation of the water balance (total discharge) of the analyzed Bokanjačko Blato catchment and its influential area as well as on the estimation of possible changes of these recharges under the influence of climate change conditions. Because there were no other relevant data for the estimation of the water balance at the test site, their analysis was performed based on the spatial distribution of specific annual discharge for the referent 30-year period (1961–1990). For this purpose, two models (Turc 1954; Langbein 1962) were used, modified, and developed for the use within the GIS environment (Horvat and Rubinić 2006). With these models, it is possible to determine spatial distribution of effective annual infiltration in the karst aquifer based on hydrogeologically determined groundwater divide and spatial estimation of the basic climatic parameters (air temperature and precipitation). Choosing of the applicable model has been performed based on the estimation of mean annual recharges (MAR) and recharges calculated by empirical regional formulas (Žugaj 1995), during which process the MAD coefficients have been compared. In the Bokanjačko Blato test area, the changes of hydrological circumstances are possible even in a shorter time scale, and for that purpose, the minimum mean monthly discharge (MMMD) has been chosen as the most suitable parameter. Since there were no measured data within the Bokanjac–Poličnik catchment, the regional study data were used (Rubinić et al., 2011). These data comprised analyses of MMMD and MAD calculated for the Gradole spring ($Q_{\text{MMM}} = 0.3206$; $Q_{\text{MA}} = 0.932$) with similar hydrological characteristics. That spring is also coastal, with karst aquifer and several flysch structures within the catchment. It is situated in Istria region, some 160 km northwest from the analyzed catchment. According to chosen model of specific discharges distribution, and based on resulting MAD coefficients, datasets of MAD were generated (MMMD was extracted from these datasets) for the relevant period of 1961–1990.

The climate change influence was estimated using methodology developed for nearby catchment of the lake Vransko Jezero (Rubinić and Katalinić 2014). In that model, the first step was generation of yearly precipitation datasets using climatological model RegCM3 (Pal et al. 2007), which has relatively optimistic scenario, and climatological model Aladin (Bubnová et al. 1995), calculated with the most unfavorable scenario of climate changes. Dataset generation was done for historical data, and for the future, up to the year 2100 (for the relevant climatological station). After the generation of the datasets, they were subjected to normalization/empirical adjustment (downscaling) against the reference historical data (1961–1990). These climatological calculations (Gajić Čapka et al. 2010) represented the basis for the hydrological calculations.

The second step comprised the connection of climatological fundamentals with hydrological circumstances in the analyzed catchment area of Bokanjac–Poličnik. Generated mean annual air temperatures and annual precipitation datasets were corrected for the relevant station, and point data from the Zadar station were applied for the whole catchment using reduction. That reduction is based on the interrelation of average annual precipitation and air temperatures values for the station and whole catchment (isothermal and isohyets map). Further data processing consists of the generation of MAD datasets according to the chosen model (Lengbain, Turc, or their combination). Resulting mean discharge value for the relevant historical data (1961–1990) was compared with the MAD value obtained from the map of specific discharge for the same period (from the isothermal and isohyets map). In that way, the reduction coefficient was calculated and applied in further calculations of MAD from generated air temperature and precipitation datasets in the whole catchment. Homogeneity of these results was tested using Wilcoxon test (Wilcoxon 1945), which was applied both, on original resulting datasets and on datasets with eliminated trend. After all, an estimation of the trend significance was provided for MAD. According to resulting annual discharge data series, synthetic arrays of the minimal average monthly discharges were determined using functional relationship (Rubinić et al. 2011). The final hydrological assessment of the climate change influences on the research area was conducted by comparison of chosen statistical parameters and their interrelations

for the three characteristic periods (1961–1990, 2021–2050, and 2071–2100).

RESULTS AND DISCUSSION

Conceptual Model

The hydrological monitoring stations are situated only on temporary streams: at the entrance and exit points of the tunnel and at the Miljašić Jaruga. Inside the tunnel, draining the Bokanjačko Blato (open wall), certain discharge additions were recorded. During the period of 1996–2009, over 40% more water discharged from the tunnel compared to the inflow of water into the tunnel, causing a significant impact on the water balance. Unfortunately, these were the only hydrological stations in this area; however, these stations served as a good indicator for hydrological characteristics and changes. The BP system is recharged by the direct infiltration of precipitation into its karst aquifer and in the wider catchment area, just partially, by the stream Miljašić Jaruga, which is the only relevant temporary stream in the catchment.

For a better understanding of the hydrogeological relations, a conceptual model is presented in the form of a schematic cross section through the catchment area, approximately perpendicular to the geological structures (Fig. 3).

The conceptual model (Fig. 3) represents a coastal aquifer with active underground contact with seawater, which has an unfavorable influence on the hydrological conditions of all extraction sites. The profile is approximately perpendicular to the geological structures and shows a typical Dinaric karst coastal terrain. In the north, the Golubinka coastal spring is a cave with groundwater outflow and with discharges that vary from below 100 to almost 500 L s⁻¹, with an average of slightly more than 400 L s⁻¹ (Žugaj 1995; Pavičić et al. 2006). When the outflow decreases to below 100 L s⁻¹, chloride concentrations quickly increase, and below 60 L s⁻¹, pumping has to stop. This scenario happens only in extreme conditions and is quite rare. The occasional occurrences of higher salinity are currently dealt with by mixing with freshwater from other sources. The Miljašić Jaruga stream and karst polje Bokanjačko Blato are in the middle of the profile, and the tunnel is projected into the profile between them. Drainage of the polje toward the Miljašić Jaruga causes significant losses during high ground-

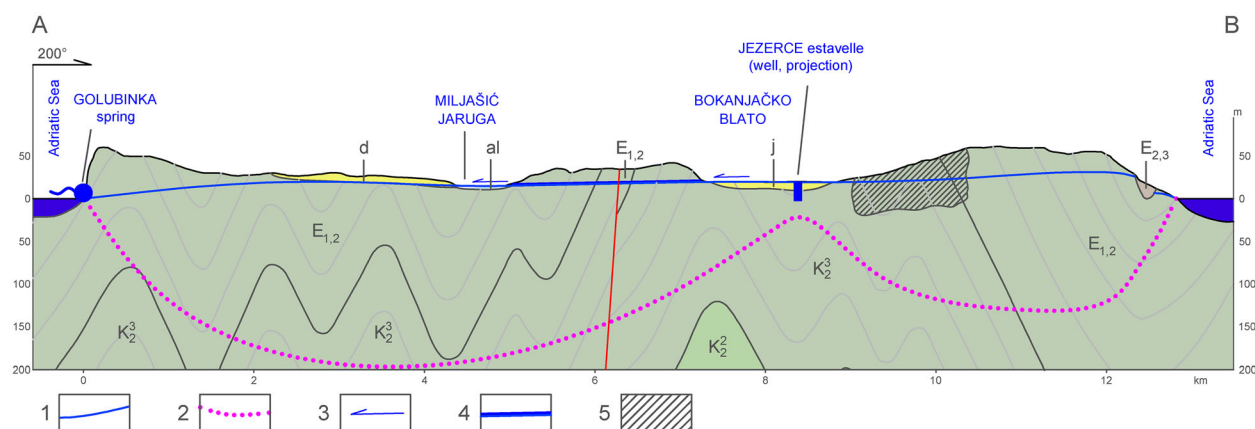


Figure 3. Conceptual model of the BP system hydrogeological relations in the form of a schematized hydrogeological cross section. Due to the mild relief, the vertical scale is exaggerated tenfold. 1-Groundwater level, 2-approximation of seawater intrusion zone, 3-periodic surface stream, 4-drainage tunnel, 5-grout curtain (projected on profile).

water levels and is followed by seawater intrusions that start to occur in the estavelle Jezerce (pit well); however, these intrusions only occur under extreme hydrological conditions and result in chloride concentrations just slightly above the level allowed. Before the tunnel construction, the polje was periodically flooded. Since the tunnel is not lined, it is open and connected with the surrounding rock mass (karstified limestones); thus, the tunnel acts as a local drain and groundwater levels are close to the tunnel. In southern part of the profile, a flysch barrier close to the coast (1) keeps the groundwater level higher than would be expected and (2) prevents (decreases) seawater intrusion toward the Bokanjac wells and Jezerce estavelle/well from the closest point.

Analysis of Selected Water Quality Indicators

Carbonate dissolution and the interaction of groundwater and seawater are the predominant influences on the groundwater quality of this coastal karst system. According to the chemical composition, the hydrochemical facies ranged from Ca-HCO₃ to a Na-Cl-type (Fig. 4), depending on the distance from the coast and the flow dynamics. The piper diagram presents the results of several hydrochemical analyses at the groundwater extraction sites. The coastal karst spring Golubinka, in the northern part of the BP catchment, was under the greatest influence of seawater, which caused the chloride to reach 7384 mg L⁻¹. The spring Boljko-vac, which is relatively close to the coast of Nin Bay,

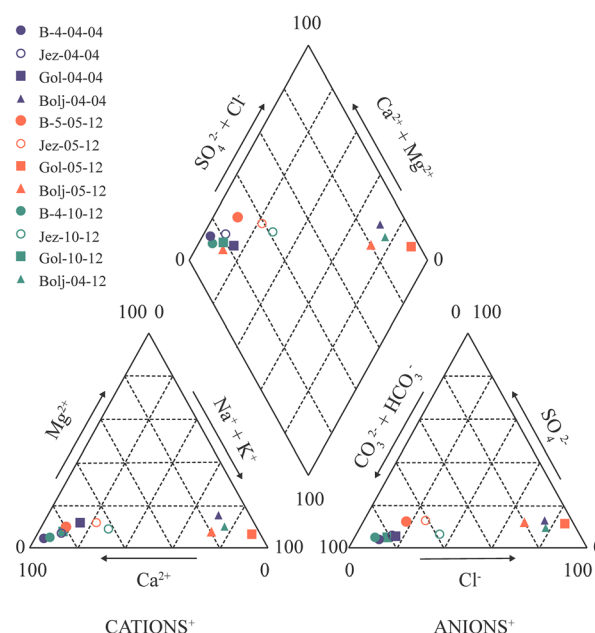


Figure 4. Piper diagram showing the hydrochemical facies of the observed springs.

was also under the influence of seawater, with a maximum chloride content of 4666 mg L⁻¹. The influence of seawater on the other extraction sites in the area of the Bokanjčko polje itself (Jezerce, Bokanjac wells) was limited to unfavorable, exceptionally long and dry hydrological conditions, which occurred rarely.

The karst aquifer was more influenced by seawater with its proximity to the coast (Fig. 3). Due to the difference in density, the denser seawater

spreads in a wedge-like form underneath the freshwater. To a certain extent, this phenomenon can be described by the Ghyben (1888)–Herzberg (1901) low in dynamic condition approximation (Bear 1972; Vacher 1988). However, the conditions in the research area are much more complex because the karst aquifer is highly heterogeneous and the model does not have a sharp interface between the two fluids, but a wide and irregular transition (or mixing) zone. Such relations depend on the groundwater pressure, and when the water table level decreases (the aquifer is unconfined at the regional scale), the seawater wedge penetrates deeper into the hinterland, and the transition zone influences reach higher zones of the aquifer. The seawater penetration is significantly enhanced in extraction zones, and the phenomenon of “up-coning” results in salinization of extracted groundwater. The only remedy in such occasions is a decrease in pumping quantity or even cessation of extraction for a certain period. In some parts of the BP system, favorable geological conditions prevent or significantly decrease these phenomena, e.g., (1) two zones of flysch rocks near the southwestern coast that act as a hydrogeological barrier, (2) similar but even more emphasized circumstances from the northeast, and (3) a combination of flysch and Quaternary soils in the zone near Boljkovac (Fig. 1). In natural conditions, only the Golubinka spring is subjected to periodical salinization, which also usually occurs with the Boljkovac well during pumping. The groundwater levels in the whole aquifer generally decreased after the 1963 construction of the hydromelioration system. Before the construction, Bokanjačko Blato was a periodical lake, but is currently drained through a network of canals and the tunnel to Miljašić Jaruga. A considerable volume of water is used for irrigation purposes. The tunnel itself acts as a drain for the nearby karst aquifer. The extraction has also increased since 1963, and these three circumstances have caused a long-term decrease in the water head and a periodic but very slight increase in the chloride concentrations in the Jezerce well (naturally estavelle). A grout curtain was constructed in the west from Bokanjačko Blato to increase groundwater pressure but with questionable results.

The two most prominent indicators of the unfavorable impacts on groundwater are the chloride and nitrate concentrations caused by excessive exploitation (chloride) and agricultural activities (nitrates) (Fig. 5). The Bokanjac wells (B-4 and B-5) were considered as a single location because they

had very similar hydrochemical parameters due to their proximity to one another. The high chloride content in the groundwater is mostly the result of the mixing with seawater, and nitrates are the best indicator of agricultural pollution due to fertilizers. Because of the draining of the former lake Bokanjačko Blato for agricultural purposes, the nitrate concentrations increased. Today, extensive agriculture activities in this area are close to the aquifer and extraction sites.

During the observed period, the analyzed chloride concentrations showed upward trends at Boljkovac, and especially at Golubinka, while the concentrations were stagnant at the other extraction sites. The upward trends were probably the combined result of natural influences (coastal karst aquifer), the negative impacts in the catchment area such as overexploitation and the sea level rise recorded at the Golubinka tide gauge station. The other monitoring sites positioned in the karst polje had stagnant chloride trends because these sites were more distant from the sea or were located behind a hydrogeological barrier that largely prevented the seawater intrusion. However, in cases in which chloride concentrations increased at sites with stagnant chloride trends, the less permeable sediments would have a negative role due to cation exchange, which slows the retreat of seawater intrusions.

Unlike chlorides, the trends of nitrate concentrations were almost stagnant at Golubinka, while at the sites of Boljkovac and at the Bokanjac wells, trends of increase were recorded (Fig. 5). In contrast to chlorides, the karstic flow conditions and rapid hydraulic response may have a positive role in nitrate concentrations because the nitrate is quickly washed out from the system. However, low hydraulic gradients, increasing pressures due to intensive agricultural activity, and adverse climate changes reduce the possibility of a quick washout from the BP karstic system. Although nitrate concentrations remained below the maximum allowable concentration in drinking water (50 mg L^{-1} , Official Gazette 125/2013), the negative nitrate trends through the period of 12 monitoring years, at all extraction wells except Golubinka, were at a very high level (approximately 20% for Jezerce and 50% at the Boljkovac spring and Bokanjac wells). The lower negative trends at Jezerce were probably the result of its location in the karst polje Bokanjačko Blato, which has a relatively thick cover of Quaternary deposits that decrease pollution of the karst

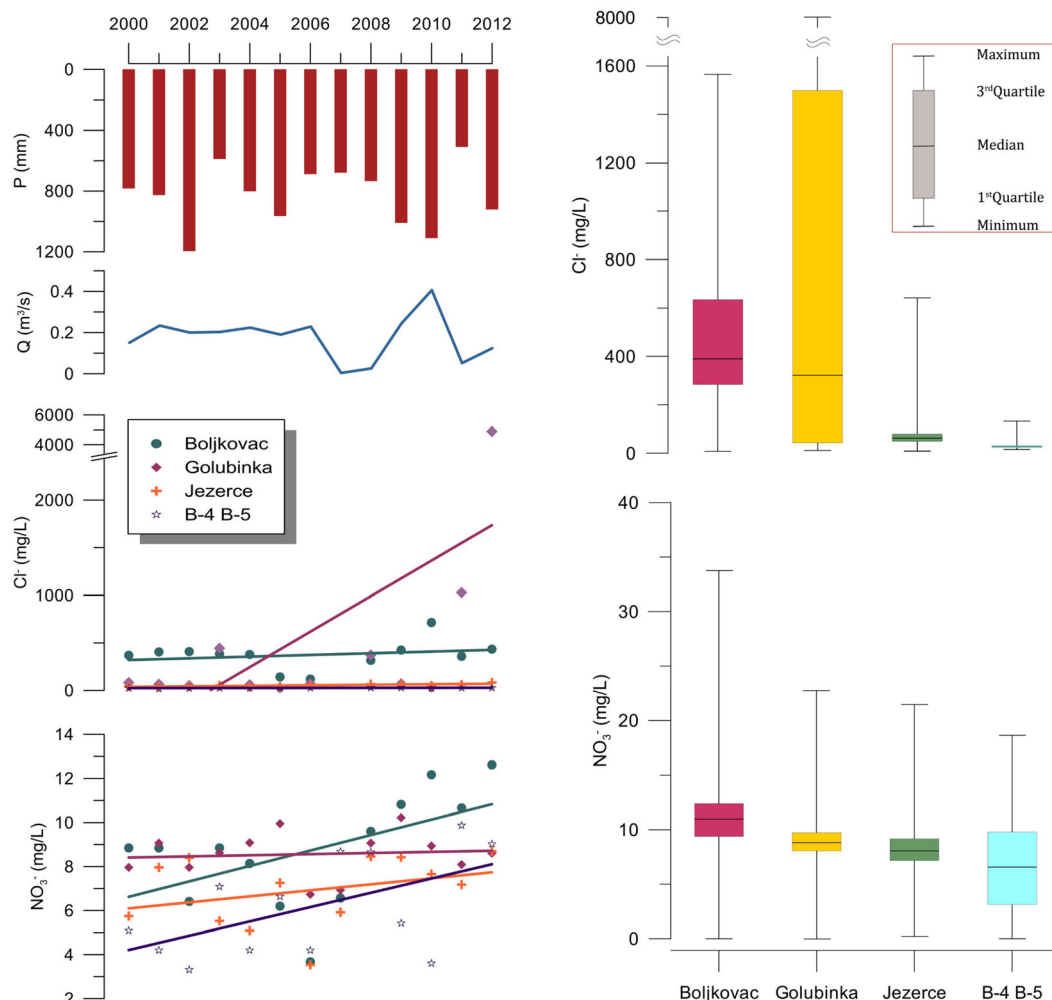


Figure 5. Annual medians of chloride and nitrate concentrations in relation to annual precipitation and discharges (left) of Miljašić Jaruga (the only stream). Box-whisker plots of chloride and nitrate concentrations (right).

aquifer below the polje through their purifying capacity and by slowing infiltration.

The water quality monitoring showed that, during the period of observation, the groundwater quality drastically deteriorated at the observed locations. Therefore, measures are necessary to prevent a further deterioration in the water quality that could lead to far-reaching consequences for the water supply in this area.

Hydrological Analyses and Assessment of Possible Impacts of Climate Change

The starting point in the estimation of the effects of climate change on water resources is the

climatic characteristics within the historical monitoring period as well as prediction of their effects in a future period. In Table 1, data are presented on the historical measurements and based on the calculations by RegCM3 and Aladin models (Gajić Čapka et al. 2010). These data were used for further hydrological modeling. The historical data are shown for the 30-year climatic period of 1961–1990, while two future 30-year periods were modeled (2021–2050 and 2071–2100). The modeling of the historical data was very similar for both, the Aladin and RegCM3 models. For the future predictions, both models predicted significant changes in air temperature ($p < 0.0001$ for RegCM3; $p = 1.0$ for Aladin for the period of 2010–2100), especially toward the end of the century. Considering precipita-

Table 1. Characteristic values that describe monthly and annual precipitation quantities and air temperatures for the Zadar climatic station based on modeling for three different periods (Gajić Čapka et al. 2010)

Model	Mean annual precipitation (mm)					Mean annual air temperature (°C)				
	Mean	SD	Cv	Max	Min	Mean	SD	Cv	Max	Min
Historical dataset 1961–1990										
Observed	916.3	179.1	0.20	1433.9	610.5	14.7	0.30	0.02	15.2	14.1
Calculated 1961–1990										
RegCM3	916.3	143.1	0.16	1245.6	572.4	14.7	0.55	0.04	15.8	13.1
Aladin	917.4	209.3	0.23	1442.6	491.5	14.7	0.47	0.03	15.5	13.5
2021–2050										
RegCM3	931.3	169.8	0.18	1381.3	630.2	15.9	0.62	0.04	17.4	14.5
Aladin	903.2	214.0	0.24	1386.6	459.0	16.3	0.55	0.03	17.8	15.3
2071–2100										
RegCM3	926.2	175.2	0.19	1328.8	574.4	18.0	0.69	0.04	19.6	16.7
Aladin	780.1	199.8	0.26	1352.2	423.3	18.0	0.57	0.03	18.9	16.6

tion, the models differed: RegCM3 predicted that precipitation at the end of twenty-first century would be very similar to that of historical data ($p = 0.641$ for the period of 2010–2100), while Aladin predicted a significant decrease in annual precipitation ($p = 0.113$ for the period of 2010–2100), followed by quite emphasized climatic extremes.

Since the model input data were connected to a single point (Zadar station), the spatial distribution was determined based on isohyet and isotherm maps for the chosen historical period of 1961–1990 and according to the proportional interrelations of their modeled changes. The specific discharges were modeled in the same manner. For the spatial distribution of the specific annual discharges, the combination of two models was used—Turc's (1954) and Langbein's (1962), which were modified and developed for GIS (Horvat and Rubinić 2006). The model selection was based on the estimation of the MAD (at the perennial level correspond to the recharge) calculated by empirical regional formulas, and during the process, the mean annual discharge/recharge coefficients were compared. In the BP catchment area, changes in the hydrological circumstances are possible even on a short time scale; therefore, the minimum mean monthly discharge was chosen as the most suitable parameter. The determination of the spatial distribution of the effective annual infiltration observed in the karst aquifer was based on the hydrogeologically determined groundwater divides and the spatial estimation of the basic climatic parameters (air temperature and precipitation).

The model calibration and validation, as well as the verification of results, could not have been carried out with the approach customarily used with more hydrologically complex models. The fundamentals were not available for such an analysis because no measured data on recharge and discharge existed for this karst system. Therefore, the calibration was performed within the two segments as follows:

- The calibration of the hydrological model's input data (spatial distribution of the average annual precipitation and air temperature) for the referent 30-year period (1961–1990) was performed by the establishment of correctional interrelations among the point data measured in the chosen station and the values achieved based on their spatial distribution.
- The second segment of the calibration performed for the same 30-yearlong dataset was chosen based on the authoritative model for the estimation of the mean annual precipitation/effective infiltration. The choice of the model was carried out by the comparison of the adjustment of the modeling results of the simplified (presented here) model and the results of a few regional empirical dependences, and based on the comparative results, the annual discharge was selected.

Results of the Turc's model were accepted as plausible. For the relevant period of 1961–1990, this model calculates value of the mean annual recession

coefficient 0.31, which is close to 0.35, result for Hellenic karst (Žugaj 1995). Within the analyzed period, resulting annual recession coefficient varied from 0.16 to 0.52, which is also acceptable in regional scale. The validation of the defined model was performed based on the results of modeling performed for the period of 1991–2009 following the relevant climatological 30-year dataset (1961–1990). This period was chosen because some extremely high values of the mean annual air temperatures were noted, accompanied by an extremely low amount of precipitation. For example, the extremely dry year 2003 was characterized by mean annual discharge of $0.453 \text{ m}^3 \text{ s}^{-1}$, with annual recession coefficient of only 0.13 (annual precipitation within the catchment was 608 mm, and mean annual air temperature was 14.8°C). Similar, but even more critical MAD values were predicted in generated data for the period of 2071–2100. For the RegCM3 model, the most critical year is 2017 ($0.435 \text{ m}^3 \text{ s}^{-1}$ —with annual recession coefficient of 0.13), while Aladin model predicts year 2090 as driest ($0.433 \text{ m}^3 \text{ s}^{-1}$ —with annual recession coefficient 0.12). The most worrying issue is not only the fact that frequencies of such dry years will be more often, but also expectation of much more frequent occurrences of very low discharge values by the end of twenty-first century—especially in the Aladin model results.

Table 2 shows the differences obtained for these results, compared to the historical referent dataset of 1961–1990. The different models produced different predictions of the changes in the balance according to the predicted climate changes. For the period of 2021–2050, at the MAR level, the expected changes (decrease) from the state in the referent climatological period of 1961–1990 were

from 8.1 according to the RegCM3 to 11.2% according to the predictions based on Aladin model. The changes in the MMMD, determined based on the methodology of the minimum and average annual flow ratios calculations, were in a similar range from 7.2 to 10.2%. In the period of 2071–2100, a drastic deterioration of the water balance was calculated. According to the calculations, the MAR would decrease between 32 and 38.5%. The changes in MMMD were even more emphasized, with decreases from 39.6 to 46.8%. Aladin predicted greater changes, with somewhat smaller changes predicted by RegCM3 model. Notably, the RegCM3 gave significantly higher minimal values, a result that suggested that the Aladin model was more reliable.

The distribution of MMMD (Fig. 6), which is the most important water supply limitation factor for the relevant time series, clearly presents that both data series have expressed statistically significant decreasing trends ($p = 0.0005$ for RegCM3, $p < 0.0001$ for Aladin). Notably, the Aladin model generated a time series of discharges with a more frequent occurrence of dry years, and such drought was never recorded in the historical datasets. Heading toward the end of this century, MMMD will increasingly and more frequently drop below 200 L s^{-1} , even 100 L s^{-1} , with a minimum of 60 L s^{-1} ; this amount of discharge is much lower than the demands on the water supply ($150\text{--}300 \text{ L s}^{-1}$).

CONCLUSIONS

It is very doubtful that the water supply from the BP karst system can be secured without planning for the influences of the predicted climate changes in a certain long-term period. During the extensive

Table 2. Changes in the percentages of the main balance parameters MAR and MMMD for selected periods

Model	Mean annual recharges (%)					Minimum mean monthly discharges (%)				
	Mean	SD	Cv	Min	Max	Mean	SD	Cv	Min	Max
Historical dataset 1961–1990 ($\text{Q}/\text{m}^3 \text{ s}^{-1}$)	1.766	0.821	0.46	0.579	4.321	0.541	0.235	0.43	0.193	1.254
2021–2050										
RegCM3	– 8.1	– 35.7	– 28.3	52.2	– 30.0	– 7.2	– 35.3	– 30.2	47.7	– 28.3
Aladin	– 11.2	– 25.8	– 15.2	– 0.9	– 29.6	– 10.2	– 25.1	– 16.3	– 1.0	– 27.9
2071–2100										
RegCM3	– 32.0	– 46.8	– 21.7	– 24.2	– 47.9	– 39.6	– 36.6	7.0	– 65.3	– 45.4
AlUadin	– 38.5	– 41.9	– 4.3	– 26.1	– 39.2	– 46.8	– 30.6	32.6	– 67.4	– 35.1

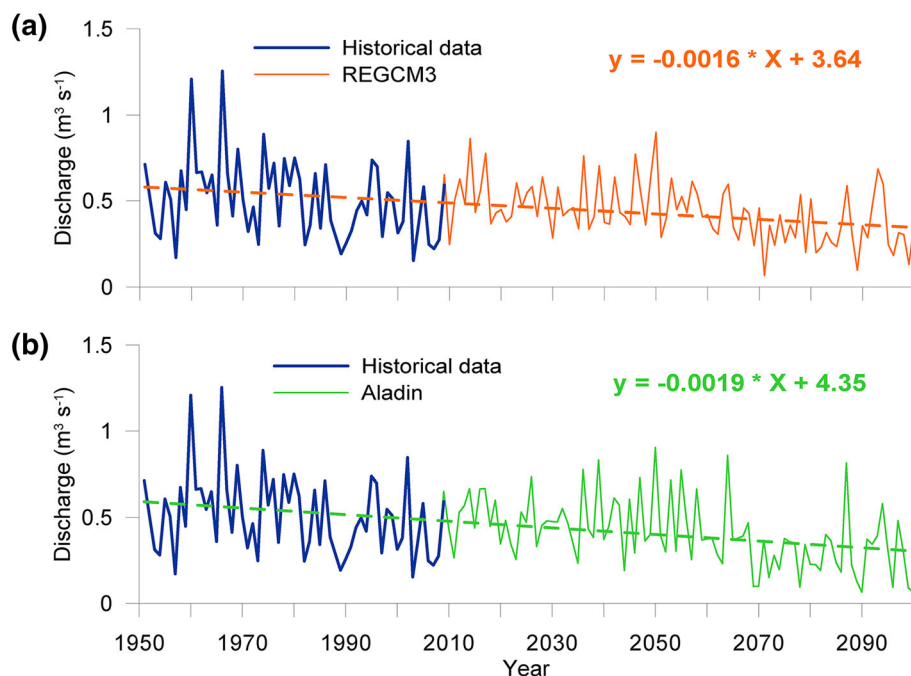


Figure 6. Presentation of time changes in MMD generated for the period of 2010–2100: (a) RegCM3 and (b) Aladin.

pumping after drought periods, saline intrusions occur even today, which limit the utility of this public water supply system.

The results of hydrological modeling showed a decreasing trend in the MAR. Because of the further deterioration in climatic conditions (increase in temperature and decrease in precipitation), years with extremely low recharge, even with historically unrecorded recharge values, could increase in frequency. The unfavorable predictions of climate changes point to a decrease in the available pumping quantities in the near future. The total annual recharges will decrease, as will the minimum mean monthly discharges. Pumping at present rates will lead to more instances of saline intrusions into this karst aquifer. The situation will only worsen over time.

In addition to water quantity, water quality is also endangered. With the drastic increase in chloride concentration trends at Golubinka and Boljkovac springs, when combined with the predicted climate changes (sea level rise, lower recharge, temperature increase), the danger is real that chloride concentrations will increase even more. At the wells with stagnant chloride concentrations

today, trends of increase are expected in the near future, but in this case, the less permeable sediments in the karst field would have a negative role due to the process known as cation exchange, which slows the advance of a saltwater intrusion but also slows its retreat. Although nitrate concentrations remained well below the maximum allowable quantities in drinking water (50 mg L^{-1}), during the monitoring period of 12 years, the increase in these concentrations was alarming (by 20% at Jezerce and by 50% at Boljkovac spring and Bokanjac wells).

The results of monitoring trends in groundwater quality showed a worrying increase in both chloride and nitrate values at all monitored locations. The combination of natural and anthropogenic influences on this extremely sensitive karst coastal system has serious adverse consequences. It is crucial that the protection of karst coastal systems becomes one of the most important tasks to preserve the drinking water resources for the future, especially in the most endangered areas such as this one. Once contaminated, aquifers are very difficult to repair to the original state; the recovery a long-lasting process and the technical measures are expensive and not highly effective. Considering the fact that the pre-

dicted water balance deficits in the driest months when the water demands are greatest, in the middle (2021–2050) and the late twenty-first century (2071–2100), will exceed decreases of 7.1–10.2% and 39.6–47.8%, respectively, further problems can be expected. Such circumstances will, without any doubt, lower groundwater levels on a regional scale, and seawater intrusions will be much more common. Therefore, the whole system will not be able to provide water supply quantities even close to recent amounts, and water management has to consider this discrepancy. Either potable water will have to be transferred from distant catchments or other methods, e.g., desalination, will have to be considered.

The risk of further long-term deterioration in water quantity and quality remains. This deterioration will be especially emphasized when a few dry years occur successively. Therefore, it is necessary to establish continuous monitoring of the karst system dynamics. The groundwater level and the groundwater quality should be monitored considering seawater intrusions (especially chloride ion concentrations, or to make the monitoring efficient, fast and cheap, electrical conductivity) and considering agricultural pressures (nitrates). The groundwater monitoring system should also be directed to the deep and shallow zones of the karst aquifer because whether deeper karstified zones influence this aquifer from more distant areas remains unclear.

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Data Availability

Data procured as described in “Materials and Methods” section, which are not directly presented in the paper, are public domain and can be procured from the corresponding author upon request.

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